Team - 11 (Laadle)

Beacon: Guiding you through campus fog

Theme A

1 Problem Statement

We aim to build a guidance platform that helps students prioritize their goals and move in the right direction through structured senior mentorship and peer-to-peer collaboration.

2 Basic Description of the Problem and Its Need

College is not short of opportunities, but students often lack focus and clarity. Between notifications, social media, assignments, clubs, and social pressure, students feel constantly busy yet distracted. Even when students identify their goals, they face difficulty in prioritization and long-term alignment.

Traditional productivity or tracking applications often create a monitoring dynamic that feels patronizing and leads to notification fatigue. Students need a system that reduces mental clutter, provides direction, and offers reliable assurance that they are progressing effectively. Our platform addresses this gap by helping students prioritize intelligently and move forward with confidence.

3 Relevance to Theme A: Attention Economy on Campus

The theme emphasizes the challenge of attention management and intentional focus on campus. Our solution directly addresses this by helping students take control of their direction rather than being overwhelmed by distractions and fragmented priorities.

Instead of adding another tracking tool, we introduce a Guided Momentum model. The system treats users as capable individuals navigating a complex campus ecosystem. It acts as a strategic partner that provides clarity, alignment, and adaptive guidance rather than surveillance or pressure. By reducing uncertainty and mental clutter, the platform improves how students think, focus, and spend their energy.

4 Proposed Solution and Basic Implementation Summary

4.1 Guided Momentum Model

Unlike traditional tracking applications that rely on monitoring and excessive notifications, our approach treats the user as a capable adult. The system provides strategic clarity and adaptive guidance instead of pressure-based nudging.

4.2 High Priority Functional Requirements

1. User Profile

Each user profile contains:

- Goals and desires of the student.
- Achievements (especially for seniors).
- Verified proof of achievements for validation.

Profiles can be viewed by other users.

2. Graph-Based Profile Matching for Advice

We use a graph database structure consisting of:

Node Types

- **User Node**: Represents students and seniors with properties such as role, availability, trust score, current level, and active load.
- **Domain Node**: Represents areas of growth such as technical skills, branch change, exams, sports, hobbies, or career objectives. Properties include name, type, embedding, and popularity score.

Relationships

- **INTERESTED_IN**: Connects students to domains with properties like priority and current level.
- **EXPERIENCED_IN**: Connects seniors to domains with properties such as experience level and years of involvement.
- **MENTORED_BY**: Directed edge from student to senior representing active mentorship.
- **CONNECTED_WITH**: Peer-to-peer collaboration between students.

Domain Normalization Free-text interests are converted into embeddings and compared using cosine similarity with existing domain nodes. If similarity exceeds a threshold, the student is linked to the existing domain; otherwise, a new domain node is created.

Mentor Matching For each student-domain pair:

- Traverse Student → Domain → Seniors (EXPERIENCED_IN).
- Filter seniors based on availability, load, level compatibility, and trust score.
- Select top k seniors dynamically based on student priority.

Peer Matching Students are matched with peers connected to the same domain and filtered by level similarity, availability overlap, and priority alignment.

3. LLM-Based Custom Response Generation

Consensus-Based Quick Reply Using a RAG approach, the system synthesizes historical database successes with advice from high-trust seniors to provide instant answers.

Predictive FAQ Generation To handle cross-questioning efficiently, the system integrates a Quick-Reply mechanism within its predictive FAQ framework. When a student submits a query, the AI immediately retrieves semantically similar past questions and high-trust senior responses (including previously answered supplementary questions) to generate a data-backed provisional response. The student is explicitly informed that this is an automated first-look answer, while the query remains open for senior verification.

Simultaneously, the AI analyzes historical interaction patterns and common stuck points to generate likely follow-up questions. When a senior responds, these predicted supplementary questions are suggested proactively, encouraging a complete, self-contained answer in a single interaction. This reduces repetitive back-and-forth and enables faster, more structured problem resolution.

Conflict Synthesis & Anomaly Detection If new advice contradicts historical trends, the system flags it as a shift from the norm and presents both perspectives to the student. Verified shifts (e.g., policy changes > threshold) trigger an explained update and the soft deletion of outdated records.

4.3 Trust Score Computation

TrustScore depends on the following factors:

$$\text{TrustScore}_s = f(\text{StudentFeedback}_s,\ \text{SuccessRate}_s,\ \text{Consistency}_s,\ \text{Alignment}_s,\ \text{VerifiedAchievements}_s)$$

- **Student Feedback:** Aggregated ratings and qualitative reviews from students reflecting the perceived helpfulness, clarity, and impact of the senior's guidance.
- **Follow-Through Success Rate:** Percentage of cases where students report measurable progress or goal completion after implementing the senior's advice.
- **Consistency Score:** Quantifies the senior's reliability and engagement based on response frequency, responsiveness time, and sustained activity on the platform.
- **Alignment Score:** Measures how closely a senior's advice aligns with high-confidence, majority-supported responses in similar queries, with proportional penalties for significant deviations.

- **Verified Achievements (Proof of Work):** Authenticated accomplishments provided during registration (e.g., branch change, high CGPA, placement, research, open-source contributions), weighted more heavily when contextually relevant to the query.

4.4 Weighted Advice Aggregation

$$Weight_i = TrustScore_i \times Similarity(StudentState, SeniorHistoryCase)$$

If advice contradicts the student's goal, similarity becomes low.

The LLM merges the top k weighted advice responses, extracts common agreements and disagreements, and generates a refined final output for the student.

4.5 Cold Start Handling

If no relevant senior exists:

- A broadcast notification is sent to all active seniors.
- Seniors can choose to respond or ignore.

4.6 Messaging Flow

- Student opens senior profile.
- Clicks "Message".
- Message appears in "Pending Requests".
- Senior can Accept, Reject, or Ignore (auto-expire).

Senior sees context including student name, year, branch, CGPA, bio, and first message preview.

4.7 Unified Query Resolution

An LLM-driven cumulative analysis clusters overlapping student queries. Seniors answer consolidated prompts, and responses are decomposed and mapped back to individual student contexts.

5 Tech Stack

- Neo4j (Graph Database)
- Django (Backend)
- React (Frontend)
- PostgreSQL (Additional User Data)
- Pinecone (RAG)

6 Minimum Viable Product (MVP)

Included

- User Profile
- Domain Node Creation + Embedding Matching
- Senior Matching (2-hop traversal)
- Basic TrustScore Calculation
- Basic RAG Quick Reply
- Synthesized Answer Generation

Future Scope

- Conflict Anomaly Detection
- Soft Deletion Logic
- Cumulative Call Clustering
- Advanced Predictive FAQ

A Appendix

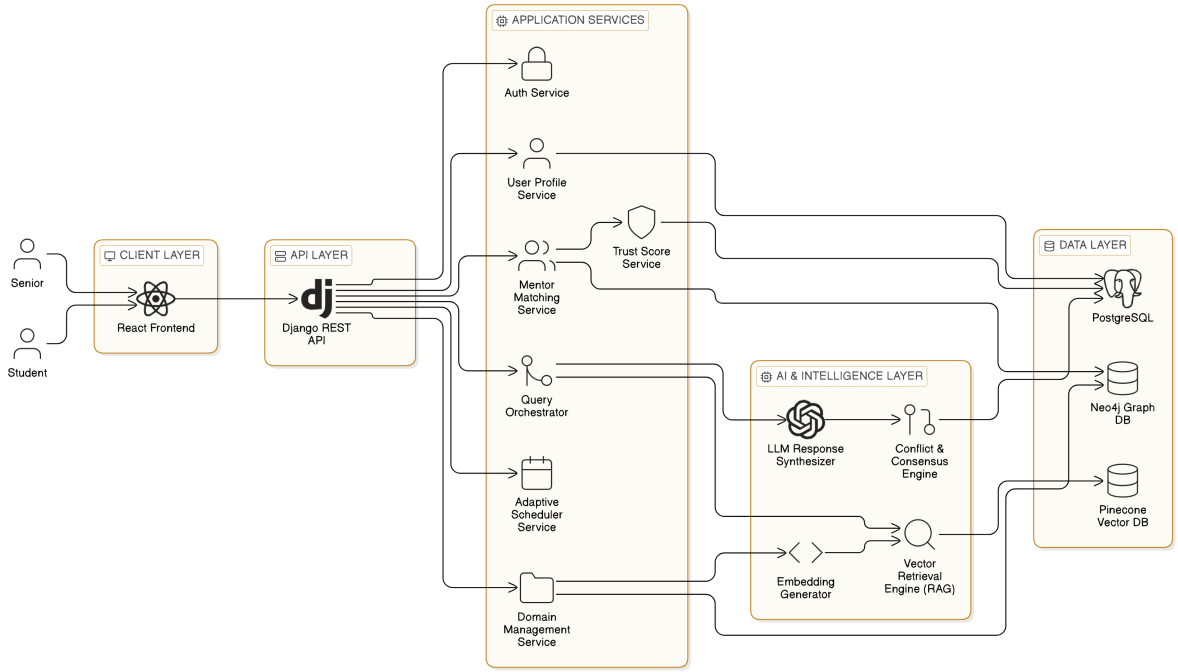

Figure 1: System Design

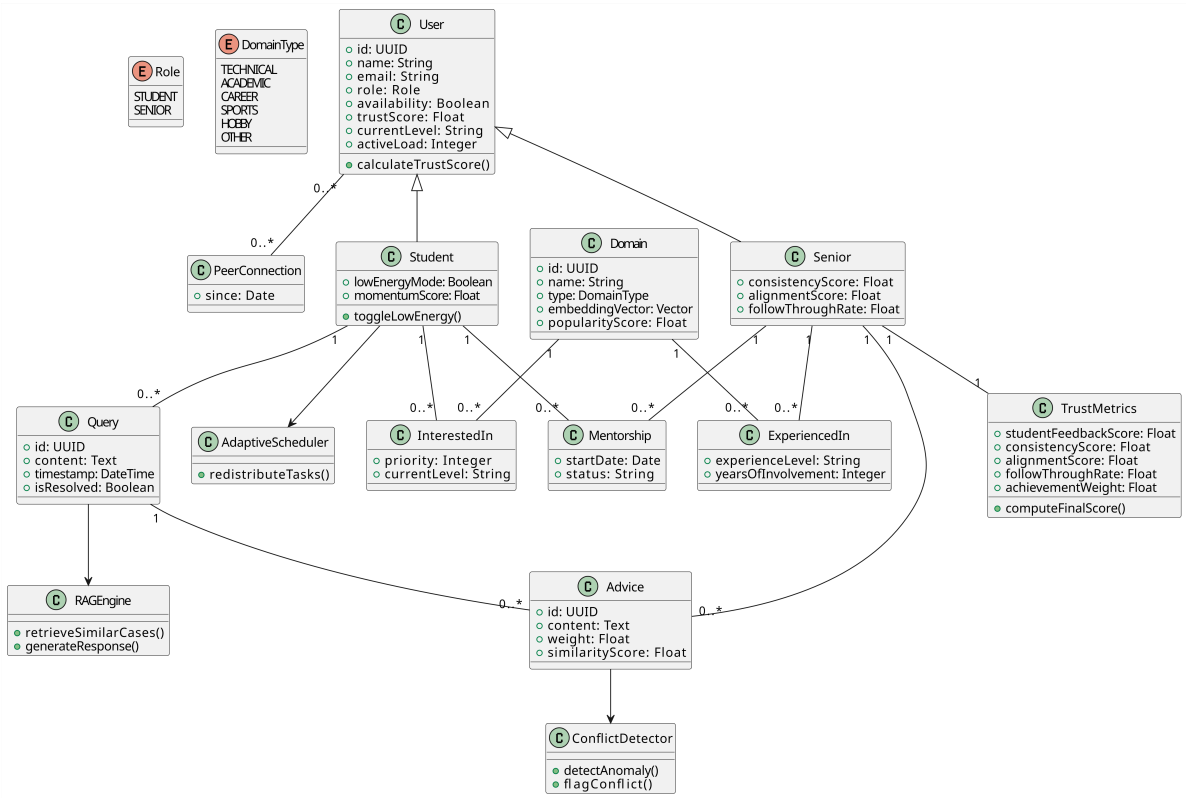

Figure 2: Class Diagram

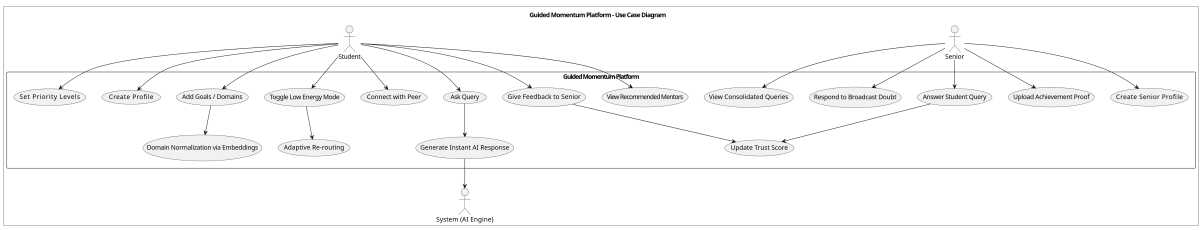

Figure 3: Use Case Diagram

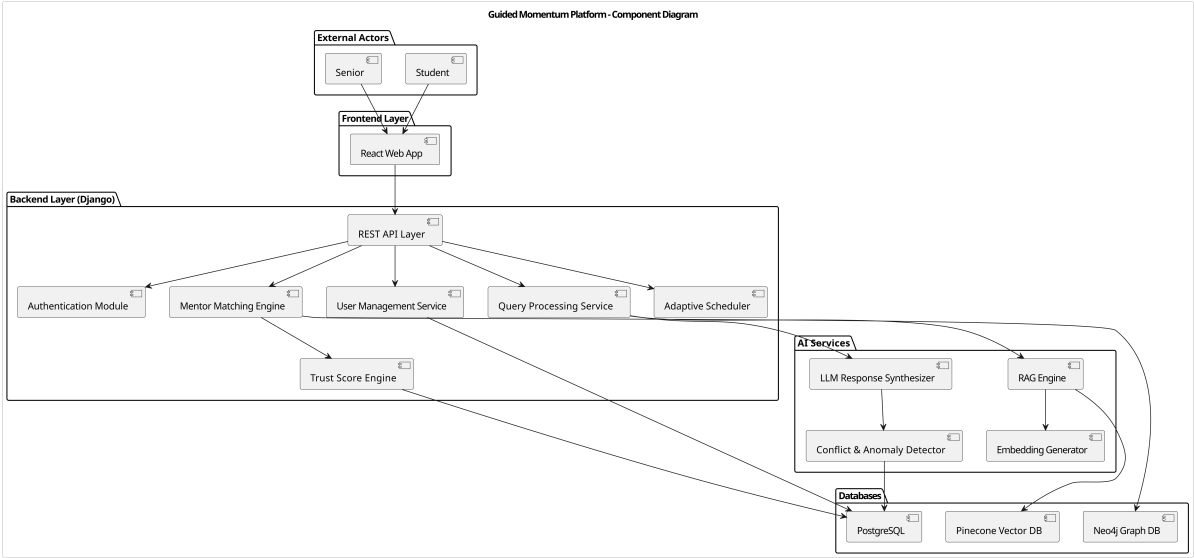

Figure 4: Deployment Diagram